
Artificial Intelligence

BS (CS) Spring 2026

Lab 07 Tasks



Learning Objectives:

1. Genetic Algorithm

Lab Tasks

Submission Instructions

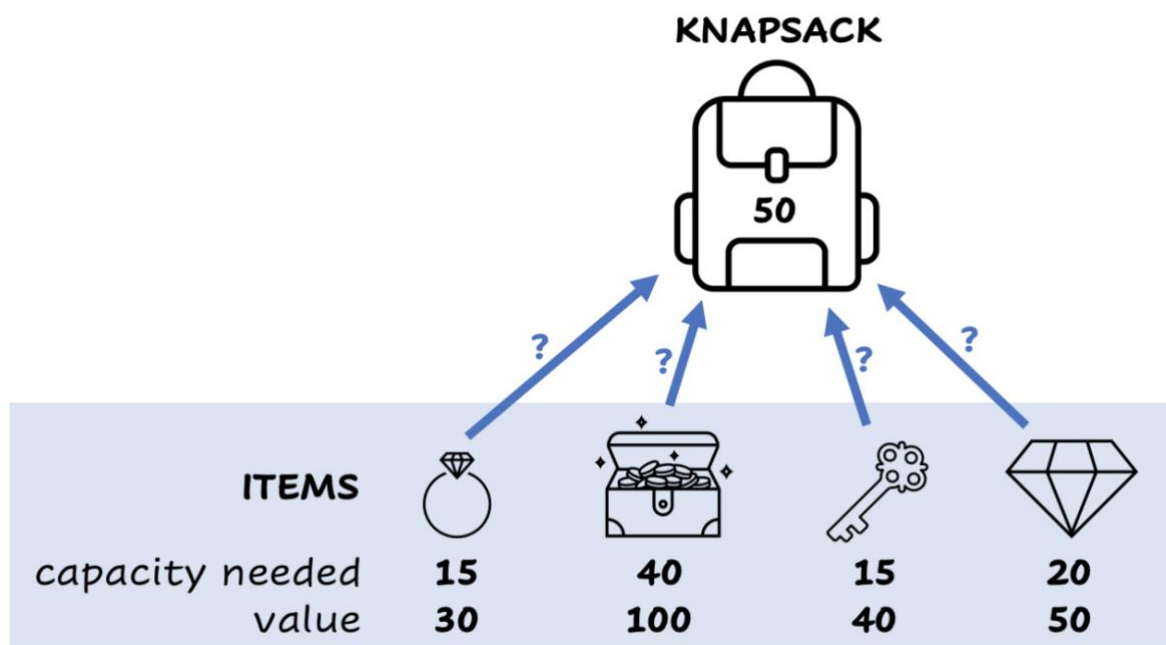
1. Create one notebook file for all questions with name *ROLLNO_SEC_LAB7_II* e.g. **i23XXXX_A_LAB07_II**
2. Submit your .ipynb file with correct name on Google Classroom under your section heading.
3. If you don't follow the above-mentioned submission instruction, you will be marked **zero**.

Plagiarism in the Lab Task will result in **zero** marks in the whole category.

Questions

The Knapsack problem is an optimization problem that deals with filling up a knapsack with a bunch of items such that the value of the Knapsack is maximized. Imagine you are in a cave full of beautiful treasures. Due to some unforeseen circumstances the cave is collapsing. You have time to fill your knapsack with treasures and then you need to run away to safety. Of course, you want to take the items with you that together bring most value. In this problem each item can be taken only once, you can either take the item or leave it. What items should you take?

The Knapsack capacity is 50. Select items so total weight ≤ 50 and value is maximum.



Solve this problem using genetic algorithm and output the results. So, we could put valuable items in the knapsack. Let's consider there are four items their weights and value are shown in figure.

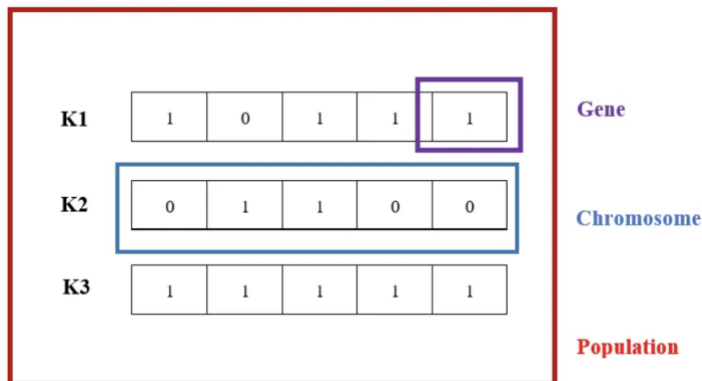
Now, we have to put all these items in a bag. Let,

Maximum Capacity in Knapsack = 50 kg

Note that, total weight should be less than the maximum weight of the knapsack.

1. Chromosomes Encoding or Initial Population:

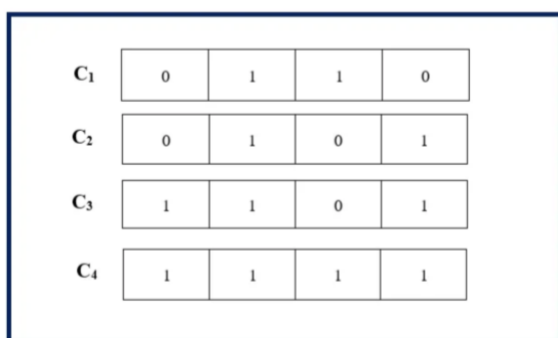
In chromosome encoding, we randomly generate an initial population of given items. In each chromosome, bits (0s & 1s) are generated randomly and each bit will represent a gene.



0 = represents the absence of an item in the knapsack

1 = represents the presence of an item in the knapsack

Generation: 1



2. Fitness Function:

The fitness function determines the ability of each individual to compete with other individuals. The individual whose fitness score is greater and weight is less than the maximum capacity of knapsack will be accepted and those whose weight is equal or greater than the maximum weight will be discarded.

For chromosome C1

	A	B	C	D
C ₁	0	1	1	0

Value of Knapsack = value of B + value of C = 100 + 40

Value of knapsack for C1 = 140

Weight of knapsack = weight of B + weight of D = 40 + 15

Weight of Knapsack for C2 = 55 kg > 50 kg

Chromosome C1 is discarded. So, from the above calculation, C1 is discarded because its *weight is larger than the maximum weight* of the Knapsack.

3. Selection:

In the selection process, the fittest individual will be selected that will be used to reproduce the next generation. On the basis of the fitness score, two parents will be selected.

e.g. Total fitness value = $C1 + C2 + C3 + C4$

e.g. Total fitness value = 51

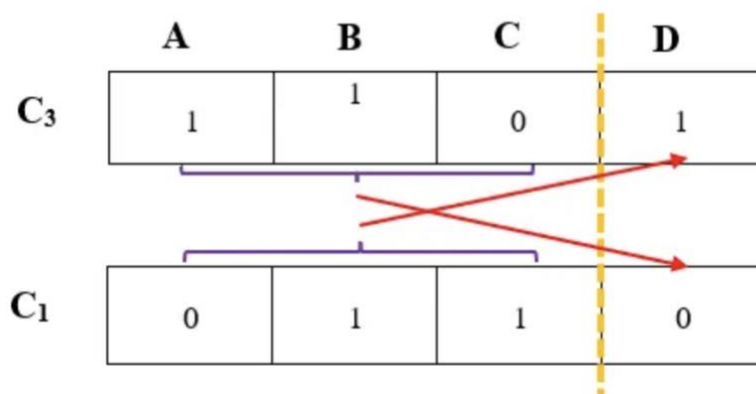
e.g. Largest fitness value is of $C3 = 24$

The individual(s) that has the *highest fitness* value will be able to produce next generation. The selection can be placed by different methods, you will choose one based upon your problem.

4. Crossover:

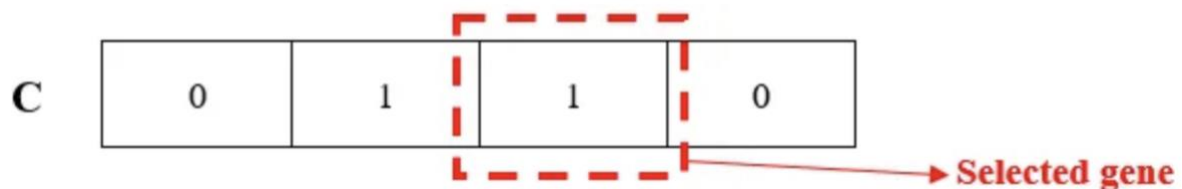
Take the selected parent's chromosomes for making and mixing the genetic material to produce an offspring.

e.g.



5. Mutation:

Introduce the diversity within the population so that the search algorithm doesn't necessarily get stuck at local maxima. Let us consider, chromosome C which will be selected randomly from the population, now randomly select a gene from C .



6. Termination (Stopping condition).